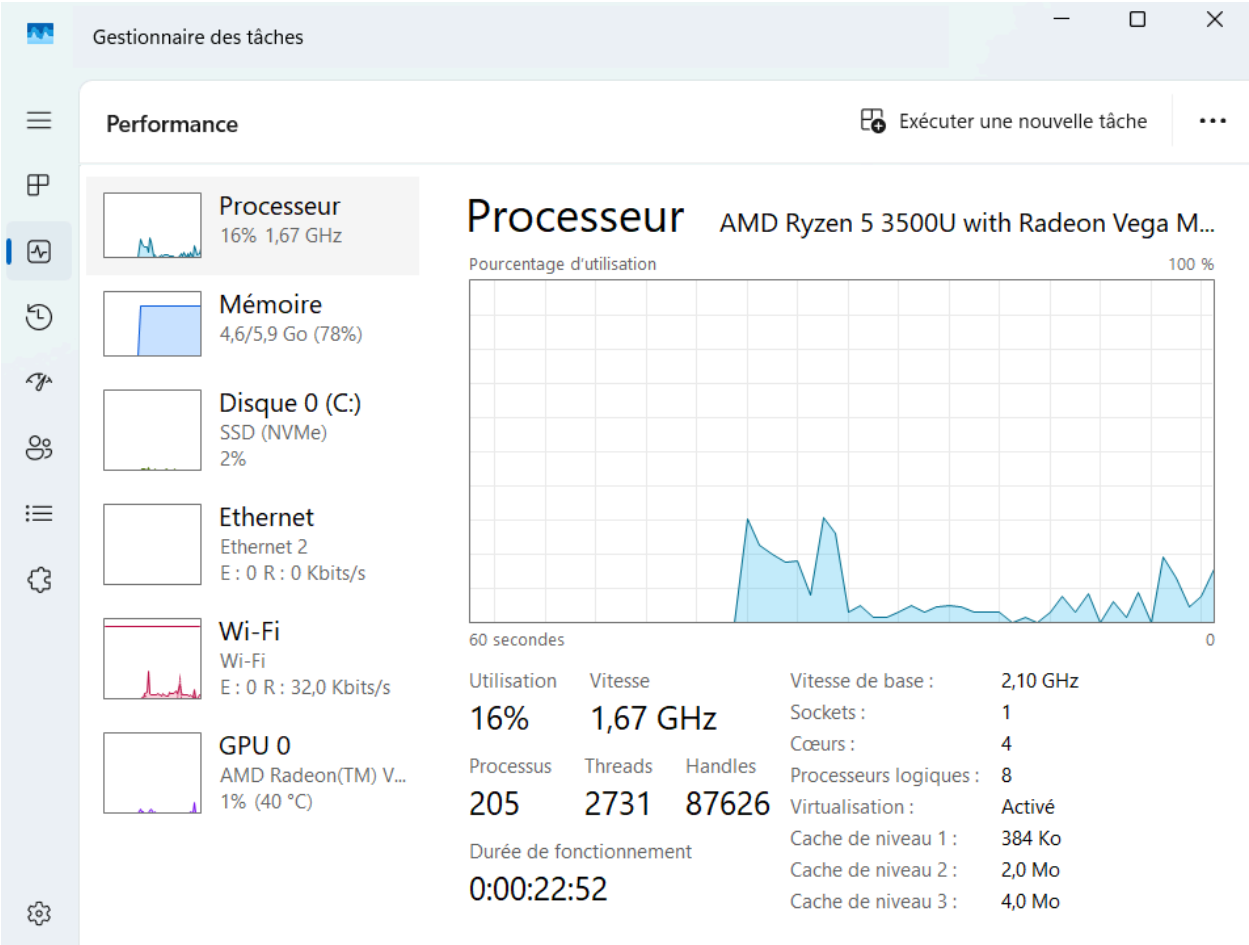
Mini-project 1: Fundamentals and Identification/Authentication

Part 1A: Setting up the Work Environment (Kali Linux VM)
Option 2 : with WSL2

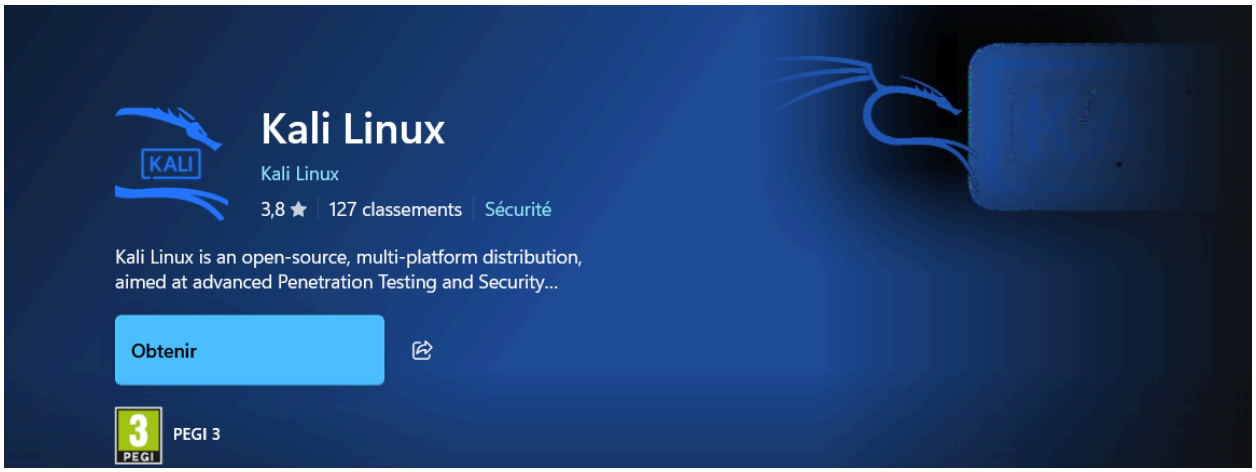


virtualisation enabled.

```
PS C:\WINDOWS\system32> Enable-WindowsOptionalFeature -Online -FeatureName Microsoft-Windows-Subsystem-Linux


Path          :
Online        : True
RestartNeeded : False
```

```
PS C:\WINDOWS\system32> dism.exe /online /enable-feature /featurename:VirtualMachinePlatform /all /norestart

Outil Gestion et maintenance des images de déploiement
Version : 10.0.26100.5074

Version de l'image : 10.0.26200.7623

Activation de la ou des fonctionnalités
[==========================100.0%==========================]
L'opération a réussi.
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> wsl --set-default-version 2
>>
Pour plus d'informations concernant les différences principales avec WSL 2, consultez https://aka.ms/wsl2

L'opération a réussi.
```




```
kali@W: ~

Installing, this may take a few minutes...
Please create a default UNIX user account. The username does not need to match your Windows
username.
For more information visit: https://aka.ms/wslusers
Enter new UNIX username: kali
New password:
Retype new password:
passwd: password updated successfully
usermod: no changes
Installation successful!
Waiting for systemd to start...
running
User account already exists, skipping creation
┌─(Message from Kali developers)
│
│ This is a minimal installation of Kali Linux, you likely
│ want to install supplementary tools. Learn how:
│ ⇒ https://www.kali.org/docs/troubleshooting/common-minimum-setup/
│
└─(Run: "touch ~/.hushlogin" to hide this message)
┌──(kali㉿W)-[~]
└─$
```

Common Steps After Installation (Valid for VirtualBox, WSL2, and Parallels Desktop) :
Check the network configuration according to the chosen option and the needs of the upcoming parts.

```
┌──(kali㉿W)-[~]
└─$ ping google.com
PING google.com (142.250.179.110) 56(84) bytes of data.
64 bytes from par21s20-in-f14.1e100.net (142.250.179.110): icmp_seq=3 ttl=116 time=12.4 ms
64 bytes from par21s20-in-f14.1e100.net (142.250.179.110): icmp_seq=4 ttl=116 time=12.8 ms
64 bytes from par21s20-in-f14.1e100.net (142.250.179.110): icmp_seq=5 ttl=116 time=17.8 ms
```

```
┌──(kali㉿W)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host proto kernel_lo
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 100
0
    link/ether 00:15:5d:b3:86:c4 brd ff:ff:ff:ff:ff:ff
    inet 172.27.70.70/20 brd 172.27.79.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:feb3:86c4/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
```

Part 1B: Implementing a Password Policy (Complexity, Renewal)

```
┌──(kali㉿W)-[~]
└─$ sudo apt update
Get:1 https://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 https://kali.download/kali kali-rolling/main amd64 Packages [20.6 MB]
Get:3 https://kali.download/kali kali-rolling/main amd64 Contents (deb) [51.9 MB]
Get:4 https://kali.download/kali kali-rolling/contrib amd64 Packages [117 kB]
Get:5 https://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [268 kB]
Get:6 https://kali.download/kali kali-rolling/non-free amd64 Packages [189 kB]
Get:7 https://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [904 kB]
Get:8 https://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [11.8 kB]
Get:9 https://kali.download/kali kali-rolling/non-free-firmware amd64 Contents (deb) [30.0 k
B]
Fetched 74.1 MB in 1min 12s (1,028 kB/s)
183 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

```
┌──(kali㉿W)-[~]
└─$ sudo apt install libpwquality-tools
Installing:
  libpwquality-tools

Installing dependencies:
  cracklib-runtime  libcrack2     libmagic1t64         libpwquality1
  file              libmagic-mgc  libpwquality-common  wamerican

Summary:
  Upgrading: 0, Installing: 9, Removing: 0, Not Upgrading: 183
  Download size: 982 kB
  Space needed: 13.2 MB / 1,025 GB available

Continue? [Y/n] y
Get:1 https://kali.download/kali kali-rolling/main amd64 libmagic-mgc amd64 1:5.46-5+b1 [338
 kB]
Get:2 https://kali.download/kali kali-rolling/main amd64 libmagic1t64 amd64 1:5.46-5+b1 [110
 kB]
Get:3 https://kali.download/kali kali-rolling/main amd64 file amd64 1:5.46-5+b1 [43.8 kB]
Get:4 https://kali.download/kali kali-rolling/main amd64 wamerican all 2020.12.07-4 [224 kB]
Get:5 https://kali.download/kali kali-rolling/main amd64 libcrack2 amd64 2.9.6-5.2+b2 [44.4
kB]
```

```
┌──(kali㉿W)-[~]
└─$ sudo apt install libpam-pwquality
Installing:
  libpam-pwquality

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 183
  Download size: 13.3 kB
  Space needed: 42.0 kB / 1,025 GB available

Get:1 https://kali.download/kali kali-rolling/main amd64 libpam-pwquality amd64 1.4.5-5+b1 [
13.3 kB]
Fetched 13.3 kB in 2s (6,628 B/s)
Selecting previously unselected package libpam-pwquality:amd64.
(Reading database ... 32110 files and directories currently installed.)
Preparing to unpack .../libpam-pwquality_1.4.5-5+b1_amd64.deb ...
Unpacking libpam-pwquality:amd64 (1.4.5-5+b1) ...
Setting up libpam-pwquality:amd64 (1.4.5-5+b1) ...
```

```
┌──(kali㉿W)-[~]
└─$ sudo nano /etc/security/pwquality.conf
```

```
# here are the per-package modules (the "Primary" block)
password        requisite                       pam_pwquality.so retry=3
password        [success=1 default=ignore]      pam_unix.so obscure use_authtok try_first_pass yescr>
# here's the fallback if no module succeeds
password        requisite                       pam_deny.so pam_pwquality.so retry=3
# prime the stack with a positive return value if there isn't one already;
# this avoids us returning an error just because nothing sets a success code
# since the modules above will each just jump around
password        required                        pam_pwquality.so pam_unix.so sha512 shadow nullok tr>
# and here are more per-package modules (the "Additional" block)
# end of pam-auth-update config
```

```
┌──(kali㉿W)-[~]
└─$ sudo nano /etc/login.defs
```

```
# Password aging controls:
#
#       PASS_MAX_DAYS   Maximum number of days a password may be used.
#       PASS_MIN_DAYS   Minimum number of days allowed between password changes
#       PASS_WARN_AGE   Number of days warning given before a password expires.
#
PASS_MAX_DAYS   90
PASS_MIN_DAYS   0
PASS_WARN_AGE   7
```

```
┌──(kali㉿W)-[~]
└─$ sudo chage -l kali
Last password change                                    : Feb 03, 2026
Password expires                                        : never
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change          : 0
Maximum number of days between password change          : 99999
Number of days of warning before password expires       : 7

┌──(kali㉿W)-[~]
└─$ sudo chage -M 90 kali
```

```
┌──(kali㉿W)-[/mnt/c/Users/Ulusa]
└─$ passwd
Changing password for kali.
Current password:
New password:
BAD PASSWORD: The password is shorter than 8 characters
```

Part 1C: Configuring SSH Authentication with Key Exchange

Étapes : 1. Générer la paire de clés SSH (sur la VM Kali - client) :

```
┌──(lea㉿kali)-[~]
└─$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/lea/.ssh/id_ed25519):
Enter passphrase for "/home/lea/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/lea/.ssh/id_ed25519
Your public key has been saved in /home/lea/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:z2CsHZ4B7PUWpyhEWN/F5Z6Z1N0II3Hh70JeA5mxIfM lea@kali
The key's randomart image is:
+--[ED25519 256]--+
|     oo   =oOo.  |
|    .o . . O.X oo|
|      + o o E + +|
|     o + o + = + |
|      o S + . O  |
|       * O o o . |
|      . + o o .  |
|            .    |
|                 |
+----[SHA256]-----+
```

2. Copier la clé publique vers le serveur (VM Ubuntu) en utilisant ssh-copy-id :

```
┌──(lea㉿kali)-[~]
└─$ ssh-copy-id kali@192.168.1.149
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/lea/.ssh/id_ed25519.pub"
The authenticity of host '192.168.1.149 (192.168.1.149)' can't be established.
ED25519 key fingerprint is SHA256:7kBTmEupa0jRIjXQ1KXy5J3u733HzhwIeLqem2r/JrA.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
    ~/.ssh/known_hosts:4: [hashed name]
    ~/.ssh/known_hosts:5: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new key
s
kali@192.168.1.149's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'kali@192.168.1.149'"
and check to make sure that only the key(s) you wanted were added.
```

3. Tester la connexion SSH par clé :

```
┌──(lea㉿kali)-[~]
└─$ ssh 'kali@192.168.1.149'
Enter passphrase for key '/home/lea/.ssh/id_ed25519':
Linux kali 6.12.25-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.12.25-1kali1 (2025-04-30) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Apr  1 00:28:26 2026 from 192.168.56.101
┌──(kali㉿kali)-[~]
└─$
```

on est bien connecté à la machine en entrant la passphrase !

4. désactiver l'authentification par mdp :

```
Fichier  Actions  Éditer  Vue  Aide
  GNU nano 8.4                                  /etc/ssh/sshd_config *
# For this to work you will also need host keys in /etc/ssh/ssh_known_host
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to "no" here!
PasswordAuthentication no
#PermitEmptyPasswords no

# Change to "yes" to enable keyboard-interactive authentication.  Dependir
# the system's configuration, this may involve passwords, challenge-respor
```

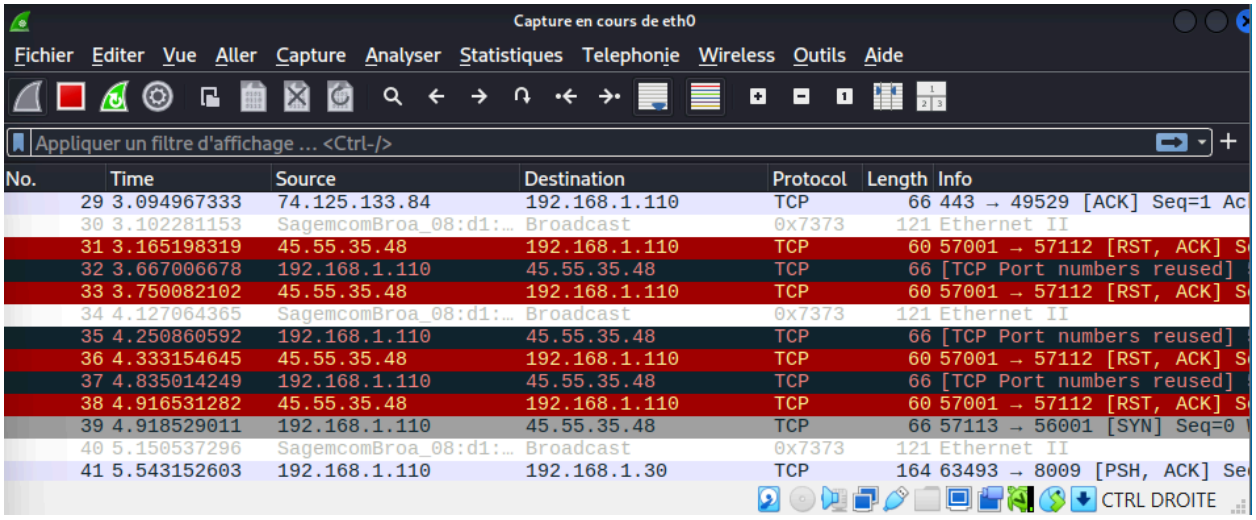
5. vérifier désactivation :

```
┌──(lea㉿kali)-[~]
└─$ ssh kali@192.168.1.149
Enter passphrase for key '/home/lea/.ssh/id_ed25519':
Linux kali 6.12.25-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.12.25-1kali1 (2025-04-30) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Apr 15 22:34:57 2026 from 192.168.1.113
┌──(kali㉿kali)-[~]
└─$ ssh -o PreferredAuthentications=password kali@192.168.1.149
The authenticity of host '192.168.1.149 (192.168.1.149)' can't be established.
ED25519 key fingerprint is SHA256:7kBTmEupa0jRIjXQ1KXy5J3u733HzhwIeLqem2r/JrA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.149' (ED25519) to the list of known hosts.
kali@192.168.1.149: Permission denied (publickey).
```

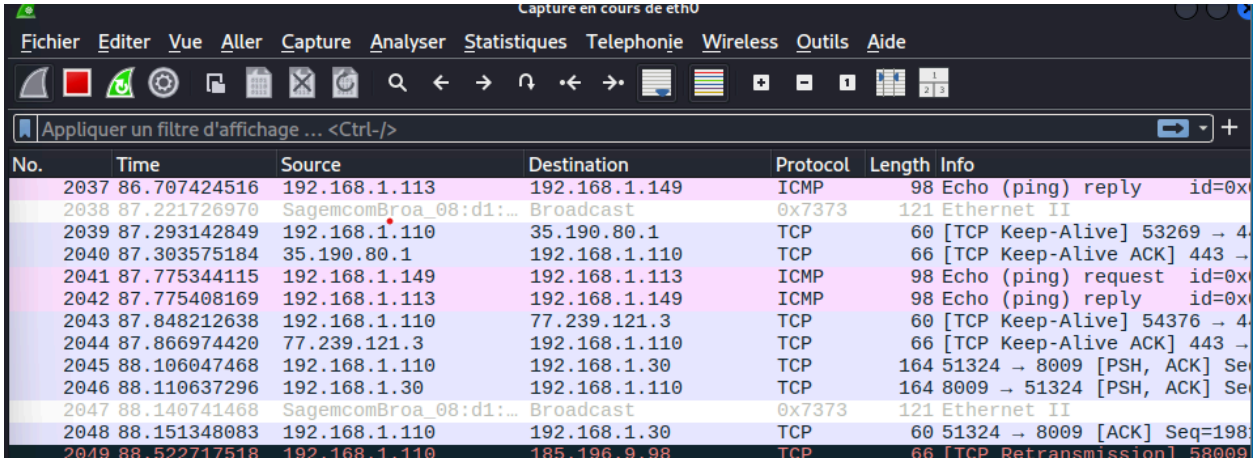
L'authentification par mdp ne marche pas mais clé ok.

Partie 1D : Introduction aux outils d'attaque et de test (Wireshark, Nmap, etc.)

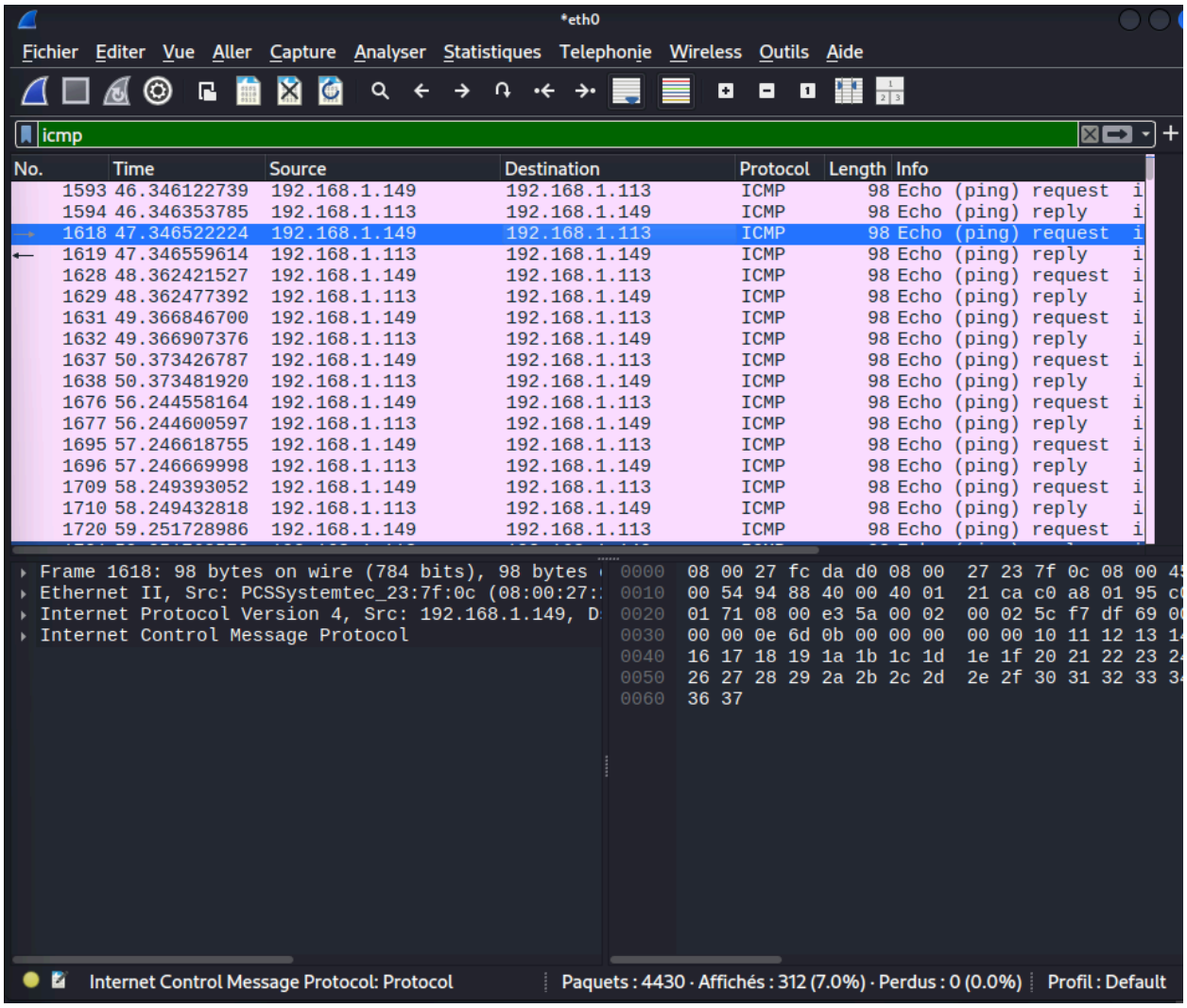
1. Wireshark - Capture et Analyse de Base (sur la VM Analyseur Kali Linux) :



captures live de eth0.



paquets ICMP ping en rose.



liste des ICMP et détail paquet request avec adresse source et destination en colonne.

2. NMAP

```
┌──(lea㉿kali)-[~]
└─$ nmap -sS 192.168.1.149
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-15 22:45 CEST
Nmap scan report for kali-001.lan (192.168.1.149)
Host is up (0.0019s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT   STATE SERVICE
22/tcp open  ssh
80/tcp open  http
MAC Address: 08:00:27:23:7F:0C (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 4.97 seconds
```

on voit les ports 22 et 80 d'ouverts.

```
┌──(lea㉿kali)-[~]
└─$ nmap -sS -sU 192.168.1.149
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-15 22:46 CEST
Nmap scan report for kali-001.lan (192.168.1.149)
Host is up (0.0028s latency).
Not shown: 998 filtered tcp ports (no-response), 998 open|filtered udp ports (no-response)
PORT   STATE  SERVICE
22/tcp open   ssh
80/tcp open   http
22/udp closed ssh
80/udp closed http
MAC Address: 08:00:27:23:7F:0C (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 9.31 seconds
```

avec ports UDP

```
┌──(lea㉿kali)-[~]
└─$ nmap -sV 192.168.1.149
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-15 22:48 CEST
Nmap scan report for kali-001.lan (192.168.1.149)
Host is up (0.0018s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 10.0p2 Debian 5 (protocol 2.0)
80/tcp open  http    Apache httpd 2.4.63 ((Debian))
MAC Address: 08:00:27:23:7F:0C (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.22 seconds
```

on voit bien le service virtualbox qui tourne.

Netcat :

```
┌──(lea㉿kali)-[~]
└─$ nc -zv 192.168.1.149 22-80
kali-001.lan [192.168.1.149] 80 (http) open
^C

┌──(lea㉿kali)-[~]
└─$ sudo tcpdump -i eth0 -n icmp
[sudo] Mot de passe de lea :
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C
0 packets captured
0 packets received by filter
0 packets dropped by kernel
```

Partie IE : mise en place de protection basique

```
┌──(kali㉿kali)-[~]
└─$ sudo iptables -L
[sudo] Mot de passe de kali :
Chain INPUT (policy DROP)
target     prot opt source               destination
ufw-before-logging-input  all  --  anywhere             anywhere
ufw-before-input  all  --  anywhere             anywhere
ufw-after-input  all  --  anywhere             anywhere
ufw-after-logging-input  all  --  anywhere             anywhere
ufw-reject-input  all  --  anywhere             anywhere
ufw-track-input  all  --  anywhere             anywhere

Chain FORWARD (policy DROP)
target     prot opt source               destination
ufw-before-logging-forward  all  --  anywhere             anywhere
ufw-before-forward  all  --  anywhere             anywhere
ufw-after-forward  all  --  anywhere             anywhere
ufw-after-logging-forward  all  --  anywhere             anywhere
ufw-reject-forward  all  --  anywhere             anywhere
ufw-track-forward  all  --  anywhere             anywhere

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination
ufw-before-logging-output  all  --  anywhere             anywhere
ufw-before-output  all  --  anywhere             anywhere
ufw-after-output  all  --  anywhere             anywhere
ufw-after-logging-output  all  --  anywhere             anywhere
ufw-reject-output  all  --  anywhere             anywhere
ufw-track-output  all  --  anywhere             anywhere

Chain ufw-after-forward (1 references)
target     prot opt source               destination

Chain ufw-after-input (1 references)
target     prot opt source               destination
ufw-skip-to-policy-input  udp  --  anywhere             anywhere             udp dpt:netbios-ns
ufw-skip-to-policy-input  udp  --  anywhere             anywhere             udp dpt:netbios-dgm
ufw-skip-to-policy-input  tcp  --  anywhere             anywhere             tcp dpt:netbios-ssn
ufw-skip-to-policy-input  tcp  --  anywhere             anywhere             tcp dpt:microsoft-ds
ufw-skip-to-policy-input  udp  --  anywhere             anywhere             udp dpt:bootps
ufw-skip-to-policy-input  udp  --  anywhere             anywhere             udp dpt:bootpc
ufw-skip-to-policy-input  all  --  anywhere             anywhere             ADDRTYPE match dst-type BROADCAST

Chain ufw-after-logging-forward (1 references)
target     prot opt source               destination
LOG        all  --  anywhere             anywhere             limit: avg 3/min burst 10 LOG level warn prefix "[UFW
 BLOCK] "
```

plusieurs règles avec ufw

```
┌──(kali㉿kali)-[~]
└─$ sudo iptables -A INPUT -p icmp --icmp-type echo-request -j DROP
```

on drop la réponse au ping.

```
┌──(kali㉿kali)-[~]
└─$ sudo iptables -L
Chain INPUT (policy DROP)
target     prot opt source               destination
ufw-before-logging-input  all  --  anywhere             anywhere
ufw-before-input  all  --  anywhere             anywhere
ufw-after-input  all  --  anywhere             anywhere
ufw-after-logging-input  all  --  anywhere             anywhere
ufw-reject-input  all  --  anywhere             anywhere
ufw-track-input  all  --  anywhere             anywhere
DROP       icmp --  anywhere             anywhere             icmp echo-request

Chain FORWARD (policy DROP)
```

on voit bien la règle de drop icmp.

5. règle ssh

```
┌──(kali㉿kali)-[~]
└─$ sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT
```

on crée la règle

```
┌──(lea㉿kali)-[~]
└─$ ssh kali@192.168.1.149
Enter passphrase for key '/home/lea/.ssh/id_ed25519':
Linux kali 6.12.25-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.12.25-1kali1 (2025-04-30) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Apr 15 22:35:26 2026 from 192.168.1.113
┌──(kali㉿kali)-[~]
└─$
```

on peut bien se connecter en ssh !

6. politique par défaut de la chaine

```
┌──(kali㉿kali)-[~]
└─$ sudo iptables -L -v
Chain INPUT (policy DROP 8085 packets, 362K bytes)
 pkts bytes target     prot opt in     out     source               destination
13487  857K ufw-before-logging-input  all  --  any    any     anywhere             anywhere
13487  857K ufw-before-input  all  --  any    any     anywhere             anywhere
 8240  374K ufw-after-input  all  --  any    any     anywhere             anywhere
 8085  362K ufw-after-logging-input  all  --  any    any     anywhere             anywhere
 8085  362K ufw-reject-input  all  --  any    any     anywhere             anywhere
 8085  362K ufw-track-input  all  --  any    any     anywhere             anywhere
    0     0 DROP       icmp --  any    any     anywhere             anywhere             icmp echo-request
    0     0 DROP       icmp --  any    any     anywhere             anywhere             icmp echo-request
    0     0 ACCEPT     tcp  --  any    any     anywhere             anywhere             tcp dpt:ssh

Chain FORWARD (policy DROP 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
    0     0 ufw-before-logging-forward  all  --  any    any     anywhere             anywhere
    0     0 ufw-before-forward  all  --  any    any     anywhere             anywhere
```

7. suppression des règles

```
┌──(kali㉿kali)-[~]
└─$ sudo iptables -D INPUT 8
```

on supprime la ligne 8 (tcp)

```
┌──(kali㉿kali)-[~]
└─$ sudo iptables -L --line-numbers
Chain INPUT (policy DROP)
num  target     prot opt source               destination
1    ufw-before-logging-input  all  --  anywhere             anywhere
2    ufw-before-input  all  --  anywhere             anywhere
3    ufw-after-input  all  --  anywhere             anywhere
4    ufw-after-logging-input  all  --  anywhere             anywhere
5    ufw-reject-input  all  --  anywhere             anywhere
6    ufw-track-input  all  --  anywhere             anywhere
7    DROP       icmp --  anywhere             anywhere             icmp echo-request

Chain FORWARD (policy DROP)
num  target     prot opt source               destination
1    ufw-before-logging-forward  all  --  anywhere             anywhere
2    ufw-before-forward  all  --  anywhere             anywhere
```

plus de ligne 8 ! bien supprimé.

```
┌──(kali㉿kali)-[~]
└─$ sudo iptables -P INPUT ACCEPT

┌──(kali㉿kali)-[~]
└─$ sudo iptables -P OUTPUT ACCEPT

┌──(kali㉿kali)-[~]
└─$ sudo iptables -P FORWARD ACCEPT
```

on remet toutes les règles à l’origine.